# Rayan Khan

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GitHub | Portfolio | LinkedIn

## FOUNDER WORK

### Sleight AI — Personal AI Entity & Automated Builder
2026–Present

Creator and builder | In development

- **Building personal AI from a user's chosen data.** Built A.L.I.C.E. as the first development case with versioned memory, source-linked evidence, owner permissions and traceable correction/deletion.
- **Trained a 267.5M-parameter conditional decoder** over frozen Gemma features: 20 epochs on 36,597 public pairs. Reloaded a 278.4M successor; its new reference/copy heads remain in progress.
- **Trained a three-stage personality candidate** for evidence representation, judgment and calibration. Reproduced all 60 saved decisions after checkpoint reload; broader behavioral validation remains pending.
- **Built FloRA's durable comparison infrastructure.** Verified 1,047 contract tests and 38 core integration cases, including restart recovery. The full behavioral comparison remains in progress. FloRA

## SELECTED RESEARCH AND ENGINEERING EXPERIENCE

### AFM Morphology Analysis Platform
Nov. 2025–Present

AI Systems Integration Engineer | Dr. Boran Ma's Research Lab

- **Shipped a researcher-facing AFM platform.** Integrated classification, segmentation and morphology analysis through Flask/Next.js, mask correction and PDF reports. Cut analysis turnaround about 65%.
- **Released a 50-image benchmark** with 90.0% classification accuracy and 0.845 mean Dice. A measured batch ran at 6.66 images/minute. Published evaluation data and scripts. Dataset

### Molecular Dynamics & Scientific Computing
2026–Present

Undergraduate Research Assistant | Dr. Sergei Nazarenko's Research Group

- **Built LAMMPS/HPC workflows for shock-wave polymer research.** Automated density profiles across six polyisobutylene blends, with Slurm batch jobs, restart handling and equilibration checks.

### Machine Learning Intern — Vision AI Development
May–Aug. 2025

- **Deployed a financial-document vision pipeline.** Reached 93–96% accuracy on 15,000 held-out documents, reduced manual review about 50% and cut inference latency over 30%.

### Open Source GPU Software
Aug. 2026–Present

External contributor | AMD ROCm Iris | NVIDIA RAFT | RAPIDS cuML

- **Merged GPU contributions into AMD ROCm Iris and NVIDIA RAFT.** Jacobi reference tests passed at 2, 4 and 8 ranks; the CUDA indexing fix includes float/double and memory-layout regression tests.

Three additional cuML PRs cover GPU data generation, IVF configuration and multiclass workflows;

## EARLY BUILDING AND LEADERSHIP

- **Deployed an automated water-reservoir controller for my community.** Adapted my robotics competition prototype to stop tank overflow in a building housing 64 families.
- **Mentored 150+ robotics participants at Roboment (2021).** Led training and project troubleshooting.

## PUBLICATIONS AND MANUSCRIPTS

**Morphology-aware classification, segmentation, and downstream analysis of Block Copolymer AFM images** — First and corresponding author. SoftwareX (under peer review); SSRN preprint available.

**Two first-author, peer-reviewed high-school papers** in the Journal of Emerging Investigators: fluorinated allopurinol (2022) and the Mpemba effect (2023).

## EDUCATION AND SELECTED HONORS

**Vanderbilt University** | B.S. Electrical and Computer Engineering | Expected May 2028 | GPA 3.94

Team Gold, International Robotics Olympiad (2022–23) | Team Silver, International Earth Science Olympiad (National Field Team round)

Gold, Queen's Commonwealth Essay Competition | Gold, The Duke of Edinburgh's International Award

Exchange student: Asia Kakehashi, Japan; Kennedy-Lugar YES, Wisconsin, USA